

DATASHEET / LIST

Safety Valve (PSV) Sizing

Computes relief area with API 520 formulas (steam/gas/liquid), then selects API 526 orifice (D–T). Tkinter input GUI.

Key Purpose

Why this program

1. Size the relief orifice by the code formulas

– API 520 area for steam / gas-vapor / liquid, then pick the API 526 lettered orifice (D-T)

→ **Effect:** The required valve size is fixed in seconds with a traceable margin, no hand calc or lookup table

2. Check vent-stack back-pressure against the limit

– ASME B31.1 App.II Fanno (steam) / Darcy (water) over the full geometry — pipe, fittings, reducer, flange

→ **Effect:** Confirms the built-up back-pressure stays under the allowable before piping is routed, avoiding valve chatter

3. Match install type and valve type to the allowable

– Open vs closed (silencer) install, and 10/25/30/50% allowable by conventional/open/bellows/pilot valve

→ **Effect:** The correct allowable rule is applied automatically so the OK/NG verdict is defensible

4. Preloaded worked sample cases

Each derived drum case fills every field on select and runs instant

Required inputs : Calc type + medium + units, then P1*/set pressure/flow, pipe ID & length, valve IN/OUT size, reducer, fitting counts, install & valve type

Main Window – Item Guide

안전밸브(PSV) 사이징 - 오리피스 / 벤트스택 배압

계산 종류: 벤트스택 배압 (1) Darcy 매체: 증기 (Fanno) (2) 단위: SI (3)

▶ 벤트스택 배압 (ASME B31.1 / Darcy)

개방형 벤트스택 (Open Vent Stack) — ASME B31.1 App.II

입력: P1* 설정압, 배관 ID/L, 밸브 IN/OUT SIZE, 리듀서, 플랜지, 수평, 용접지, 용구

샘플 (엑셀 4케이스): HP Drum (포) (5)

조건 · 배관

P1* 밸브출구 임계압	174.7	psia
설정압 Set	80	barg
유량 q (물)	20	m³/hr
토출/벤트 배관 ID	0.026	in
배관 길이 L	0.335	m
마찰계수 (인간=자동)	-	-

벤트 SIZE · 리듀서 · 플랜지

밸브 INPUT(인렛) SIZE	2.5	in
밸브 OUTPUT(아웃렛) SIZE	4	in
리듀서 대경(없으면 0)	0	in
리듀서 소경	0	in
리듀서 각도	45°	°
플랜지 L/D	0.994	-
사일런서 ΔP(배압형)	0	bar
☐ 출구손실(K=1) 포함		
☐ 과열증기(k=1.3)		

파형 수량 (ea)

90° LR 엘보 (L/D=20)	1
90° SR 엘보 (L/D=30)	0
45° 엘보 (L/D=16)	0
90° 마이터 (L/D=60)	0
45° 마이터 (L/D=15)	0
티(런) (L/D=20)	0
티(문기) (L/D=60)	0
게이트밸브 (L/D=8)	0
체크밸브 (L/D=100)	0

설치형식 · 밸브타입 (허용배압)

설치형식: 개방형 (Open vent stack) 밸브타입: (9) 개방형 반력기준 (25%)

계산 (10) CALCULATE

[벤트스택 배압] 증기(Fanno) 단위: SI
f=0.017 피팅={ 'LR90°' : 1.0 } 리듀서K=없음 밸브 IN/OUT=2.5/4.0" 설치=개방형
[벤트라인 참고] 벤트 ID=12.0" 길이=20m 엘보=6개 — 개방형은 gap 하... 배압 미반영
전체 SL/D = 24.27 f·SL/D = 0.413 P/P* = 1.537

▶ 배압(총) = 253.9 psig (17.50 barg)
▶ 허용배압(25%×설정, 개방형 반력기준 (25%)) = 290.1 psig (20.00 barg)
▶ 판정 : OK (배압 < 허용)

1. Calc type: orifice sizing or vent-stack back-pressure
2. Medium select (steam Fanno / water Darcy / gas / liquid)
3. Units select (SI / US)
4. Open vent-stack schematic with input legend
5. Sample case picker (fills all fields)
6. Conditions & pipe: P1*, set pressure, flow, ID, length, f
7. Valve size, reducer & flange; exit-loss / superheat options
8. Fitting quantities per type (elbows, tees, valves)
9. Install type & valve type (allowable back-pressure)
10. CALCULATE
11. Result: back-pressure vs allowable, OK/NG verdict

Overview

What it does

1. Computes relief area with API 520 formulas (steam/gas/liquid)
2. Then selects API 526 orifice (D-T)
3. Tkinter input GUI

Category: Datasheet / List

Folder: DATA_SHEET/SAFETY_VALV_SIZING

Main scripts: psv_gui.py

Scripts / Status: 3 scripts · Operational

Tags: API 520 · API 526 · PSV · Tkinter

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: Datasheet / List.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: `psv_gui.py`.
- Output samples and the program window above reflect the current build.

Thank You

Safety Valve (PSV) Sizing · Plant Engineering Total Service System